

JAMES PAOLO RILI

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OVERVIEW

Upcoming BCIT Computer Systems graduate and UBC Physics alumnus with 16 months of software engineering co-op experience. Proven ability in building modular software and optimising SQL queries for high-volume databases. Seeking full-time roles starting February 2027.

SKILLS

Programming Languages	Python, TypeScript, Java, C, Go, Rust, Elixir
Backend	REST, Express.js, Mongoose, PostgreSQL, MySQL, MongoDB
Frontend	React.js, Jetpack Compose, Qt
Cloud	AWS (Lambda, EventBridge, AuroraDB), Cloudflare (Pages, Workers, R2)
Tools & Methodologies	Git/GitHub, Linux, Agile, Atlassian Suite (Jira, Confluence)

WORK EXPERIENCE

Software Developer Co-op, Reporting Services Caracal Technologies <i>Vancouver, BC</i>	May 2025 - Dec 2025
- Engineered new features for an application processing high-volume (~10 million transactions/mo) transit data, delivering automated reports for clients using Python via AWS Lambda, EventBridge, and AuroraDB, as well as TypeScript and React.js. - Optimised 50+ SQL queries to fix performance issues, reducing data fetching times from over 3 minutes to under 10 seconds. - Streamlined workflow by configuring pipelines in BitBucket, writing Bash scripts, and documenting standard procedures in Confluence, reducing the risk of human error during deployment.	
RF Software Engineer Co-op, Hardware Validation NETGEAR, Inc. <i>Richmond, BC</i>	Sep 2022 - Apr 2023
- Developed and optimised Python-based software suite for WWAN hotspot device validation, ensuring that products are in compliance with the 3GPP standards while accelerating validation cycles by 50% through automation. - Built a Python library responsible for interfacing with a temperature chamber via RS-232, enabling automated temperature tests and minimising risk of human error during testing. - Added data visualisation with Matplotlib for live signal measurements, allowing testers to quickly analyse measurements and immediately pinpoint anomalies during validation.	
Physics Teaching Assistant University of British Columbia <i>Vancouver, BC</i>	Jan 2024 - Apr 2024
- Conducted weekly workshops for 50 undergraduate biology students to reinforce understanding of complex problems. - Delivered concise lectures about waves and hydrostatics to guide students' intuition of physical concepts in biology.	

PROJECTS

OTP-DigitalTwin BCIT + Iron IoT (https://github.com/ironiot/otp-dt)	Sep 2026 - Present
- Architecting a modular, fault-tolerant system leveraging Elixir's OTP framework to emulate a digital twin with data from TransLink's GTFS feed. - Engineering an infrastructure proof-of-concept and conducting analyses to assess Elixir's scalability, resilience, and integration capabilities against a Golang-based stack.	
Iron-OBID2 BCIT + Iron IoT (https://github.com/ironiot/iron-obid2/)	Apr - May 2026
- Led a 5-person agile team in building a MVP of cloud-based OBD-II dashboard system with server-based data persistence. - Compiled documentation outlining system architecture, setup, and operating procedures in order to ensure a seamless handoff to the client and enable future development teams to easily maintain, scale, and troubleshoot the system. - Wrote unit-testing with Python and automation with Bash scripts, accelerating deployment while maintaining bug-free software.	
Crustracer Personal (https://github.com/jprili/crustracer/)	Jun 2026
- Wrote a CPU-based path tracer as a starting point for a future graphics + physics engine project. - Ported the literature source code from C++ to Rust to learn about the language's memory-safe mechanisms and best practices. - Reduced raw image output size by ~66% by encoding colours in binary instead of ASCII for PPM file specification.	

EDUCATION

Computer Systems Technology Diploma, Programming Paradigms British Columbia Institute of Technology	Sep 2024 - Dec 2026
Bachelor of Science in Physics University of British Columbia	Sep 2019 - Apr 2024